

HATCHSIDE · REQUIREMENTS

HatchSide requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/hatchside		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; status of every requirement from HTS-CAL-001; safety section
0.2	2026-10-03	Amish Chadha	TRL 2; targets firmed up, R13 and R14 added (HTS-DDR-001, D10); budget as a value-engineering target
0.1	2026-09-30	Amish Chadha	Initial scaffold

HatchSide requirements

Thirteen of the fourteen requirements are met on paper by the constructable design (HTS-CAL-001); R11, corrosion resistance, cannot be shown on paper and waits for a TRL 4 coupon test. Four results rest on estimates (R5, R7, R8 and R10's design depth) and are marked so. Every requirement is verified by test at TRL 4 or later.

Table 1. Requirements and their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later)	TRL 3 result (HTS-CAL-001)	Status
R1	No entry needed for any task	All heads are fitted, lowered, worked and recovered from outside the opening	Task walkthrough for every head at a mock manhole	Every head is fitted, worked and recovered from the rim	Met on paper
R2	Fits common openings	Straddles openings from 0.5 to 1.2 m (20 to 47 in) across	Setup trials on sample manholes and hatches	Feet on a 3.0 m circle; rim guard inner face 672 mm from the centre	Met on paper
R3	Working depth	At least 10 m (33 ft) of rope below the rim	Winch drum capacity check and depth trial	15 m of rope; drum capacity to confirm on purchase	Met on paper
R4	Working load	Safe working load 150 kg (330 lb) at the head interface, proof-loaded to 1.5 times	CalRig proof-load test	Leg buckling margin 5.5 at proof with jerk; rope safety factor 14.3	Met on paper
R5	Fast setup	Two people set up in 10 minutes or less	Timed setup trial	9.0 minutes	Met on paper (estimate)
R6	Portable	No single packed load over 25 kg (55 lb)	Weigh each packed load	Heaviest load 23.7 kg (grab head)	Met on paper
R7	Quick head change	Head change in under 2 minutes without tools	Timed changeover trial	One screw-pin shackle, about 1 minute	Met on paper (estimate)
R8	Grab clears silt	At least 10 L of wet silt per cycle	Trial in a silt-filled test chamber	10.8 L at 70 % fill	Met on paper (estimate)
R9	Sampler takes depth samples	Sample centre at a chosen depth within 0.2 m	Trial in a layered test tank	Counter reads 0.1 m; tube centre 420 mm below the shackle pin	Met on paper
R10	DomeReach reaches the digester floor	Tool reaches 2.5 m below the opening rim (design depth, HTS-DDR-001, D7)	Trial at a partner digester	Hinge 2.6 m, tool 3.3 m below the rim	Met on paper
R11	Corrosion resistance	No structural corrosion after 100 hours of salt spray or sewage immersion	Material coupon test	Galvanized steel, 6082 aluminium, A4 fasteners and rope	Not verifiable on paper
R12	Value-engineering target	Frame, winch and five heads near USD 5,000	Costed bill of materials	Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 2,838 (USD 2,162 under the target)	Under the target

R13

ID	Requirement	Target	Verification (TRL 4 or later)	TRL 3 result (HTS-CAL-001)	Status
	Heads pass a small manhole	Every head, as it hangs, passes a 560 mm opening with 20 mm to spare	Pass each head through a 560 mm ring	Largest plan diagonal 525 mm (grab); DomeReach passes folded	Met on paper
R14	Stable in use	The tripod stays stable with the rope 10 degrees off vertical at the safe working load	Tilt-pull test during proof loading	Tips at 25 degrees at the safe working load	Met on paper

■ Assumptions

- Most routine sewer, tank and digester tasks can be done with grab, rake, sampler and pole tools from the rim.
- Crews carry and use a certified gas detector alongside the kit.
- A commodity two-speed hand winch with an automatic load brake is available in the first region.
- Household biogas digesters have an opening of at least 560 mm and a depth of about 2.5 m below the rim; both are to be measured with the first partner.

■ Safety

Safety: R4 and R14 are material-handling requirements for tool heads. Nothing in this document qualifies HatchSide for lifting, lowering or rescuing a person, and the winch must never be used for that. A proof load to 1.5 times the safe working load with a competent person present (CalRig) comes before any use over a real opening. Gas testing, traffic guarding and the no-entry rule apply to every trial.